Decoding Video-Elicited Emotions from EEG*

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Abstract

We address 6-class emotion classification from 31-channel EEG, with test trials drawn from a held-out session unseen at training. The setting is constrained by 336 training trials, variable trial length, large session-to-session shift, and an unknown channel montage. Our pipeline combines per-channel spectral features (band power, differential entropy, Hjorth) with per-session robust normalization, test-time CORAL alignment, and a parallel Riemannian tangent-space head. The two heads are combined via a log-probability ensemble and decoded with a hard balanced assignment (Hungarian) that exploits the known balanced-class prior. The pipeline reaches 0.652 ± 0.135 mean LOSO accuracy ($4\times$ chance), with CORAL contributing the single largest improvement ($+0.080$).

1 Raw signal visualization

ERP. For each class we average the signal across a 7-channel subset and across all trials. All six classes converge to a near-flat trace centered around -0.35 with overlapping SEM bands. ERP averaging only recovers components phase-locked to stimulus onset; for sustained affective states, ERP features are uninformative.

Time-Frequency. We compute per-trial spectrograms (Welch, 512-sample window, 50% overlap) on the channel-averaged signal and average in dB across trials. Figure 1 shows class-discriminative power below $\sim$10 Hz, dominated by 1/f drift; mid-band streaks (*anger*, 10–25 Hz) and broadband vertical bursts (*neutral*, *disgust*) indicate residual EMG and motion artifacts. This motivates high-pass filtering, edge cropping, and per-window normalization.

1.1 Signal quality audit

We check signal quality at three levels.

Trial level (train vs. test). Trial duration matches between train and test ($p = 0.39$, two-sample KS), but mean channel std, peak-to-peak amplitude, and extreme-sample fraction differ significantly ($p < 0.005$). The amplitude-scale shift exceeds typical inter-session variability, motivating test-time CORAL alignment.

Channel level. Channel 30 has mean std an order of magnitude above the rest with near-zero high-frequency content – low-frequency drift, not EMG. We exclude it. Mid-array channels (5-8, 23-26) show elevated HF ratios consistent with frontotemporal EMG; we flag but retain them, since dropping them did not improve LOSO.

*Submitted as part of the Alljoined Machine Learning Researcher take-home challenge, April 2026.

Preprint.

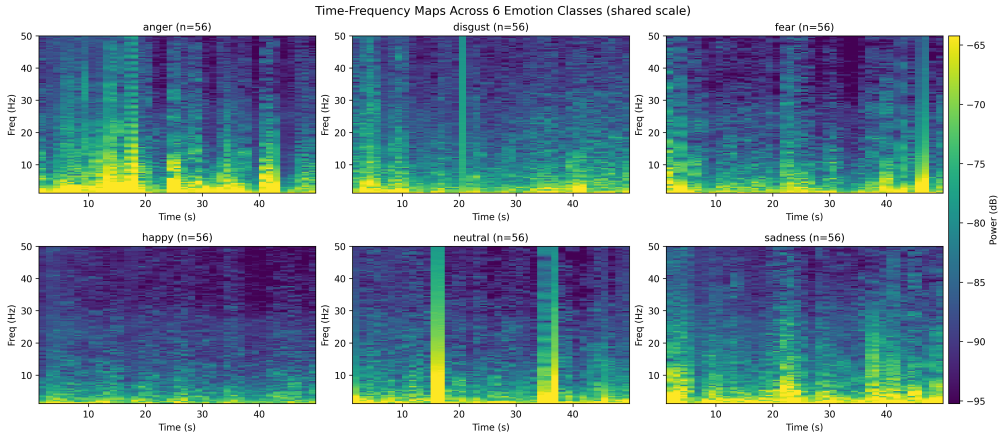


Figure 1: Trial-averaged time-frequency maps, raw data (shared dB scale).

Session level. For each training trial we compute three summary statistics: cross-channel mean of per-channel std, max peak-to-peak across channels, and the fraction of samples whose robust z-score exceeds 5. Each is itself robust z-scored across the training set, and the trial-level anomaly score is the sum of the positive parts:

$$a_i = \sum_{m \in \mathcal{M}} \max\left(0,\ \frac{0.6745\,\left(s_{i,m} - \operatorname{med}_j\, s_{j,m}\right)}{\operatorname{MAD}_j\, s_{j,m}}\right), \quad \mathcal{M} = \{\text{mean_std, max_ptp, extreme_frac}\}. \tag{1}$$

Most sessions concentrate near zero; sessions 11 and 13 are clear outliers, with four trials in session 13 exceeding 1000.

2 Cleaning and preprocessing

For each trial we (i) drop channel 30; (ii) crop to the usable signal length, then trim 3 s from the front (emotion-onset latency) and 1 s from the end; (iii) linearly detrend per channel; (iv) apply a 4th-order zero-phase Butterworth high-pass at 0.5 Hz; (v) compute a per-session robust z-score (median / MAD) per channel.

What improved. The cleaned ERP centers at zero (drift confirmed as the prior offset). The cleaned TF maps (Figure 2) lose the broadband vertical bursts; the low-frequency band reflects neural delta/theta rather than 1/f drift. Channel std becomes uniform ($< 2\times$ variation, vs. $50\times$ before). Per-session anomaly scores shrink by $\sim$75$\times$.

What remains. Three issues persist. (1) Per-session z-scoring fails to close the train-test KS gap: three of four metrics remain significantly different ($p < 0.001$), so the residual shift is structural rather than scalar – this motivates explicit feature-space alignment. (2) ERP remains uninformative; we commit fully to spectral features. (3) Short (< 2 s) filter-edge transients persist, so feature extraction operates on overlapping 8 s windows starting after the first 2 s.

3 Approach

The audit prescribes four design requirements: (a) features must be invariant to channel ordering and trial length; (b) the pipeline must close the residual train–test gap; (c) capacity must fit $\sim$312 trials per LOSO fold; (d) classifier outputs must be calibrated for log-probability ensembling.

Spectral features. On every cleaned trial we tile overlapping 8 s windows (4 s hop) and per channel compute three descriptors: log band power across five bands (delta 1–4, theta 4–8, alpha 8–13, beta 13–30, low gamma 30–45 Hz) via Welch's method; differential entropy [1], estimated as $h_b = \frac{1}{2}\log(2\pi e\sigma_b^2)$ from the same Welch PSDs; and Hjorth activity, mobility, and complexity.

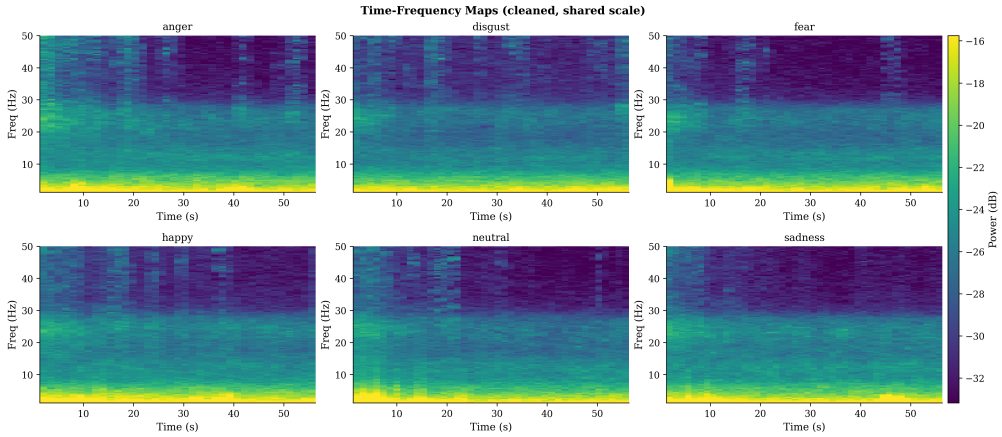


Figure 2: Time-frequency maps after cleaning. Broadband vertical bursts are gone and class-specific low-frequency structure is more visible.

Window-level features are aggregated per trial via concatenated mean, std, and median, yielding a 1170-dimensional descriptor.

Per-session normalization and CORAL alignment. Each feature dimension is z-scored within session: $\tilde{x} = (x - \mathrm{med})/(1.4826\,\mathrm{MAD})$. To close the residual structural shift we apply CORAL [2] at inference: with source covariance C_s and target covariance C_t, source features are whitened then re-colored as $X_s' = (X_s - \mu_s)C_s^{-1/2}C_t^{1/2} + \mu_t$. CORAL is unsupervised and adds no parameters; in LOSO it is applied per fold using only the held-out session's features.

Spectral classifier. A multinomial logistic regression with ℓ_2 regularization is trained on the aligned features. The strength C is chosen by 3-fold stratified inner CV over $\{0.01, 0.1, 1, 10\}$. We use LR rather than SVM because the downstream ensemble combines log-probabilities, and LR's softmax is inherently calibrated.

Riemannian tangent-space head. A parallel head operates on raw 4 s/2 s windows: each window's 30×30 sample covariance is estimated via OAS shrinkage [3], projected to the tangent space at the Riemannian mean, and fed to a logistic regression. Trial-level probabilities are obtained by averaging window-level probabilities.

Log-probability ensemble. The two heads are combined as

$$\log p_{\text{ens}}(c \mid x) \propto w \log p_{\text{spec}}(c \mid x) + (1 - w) \log p_{\text{riem}}(c \mid x), \tag{2}$$

with w selected on LOSO.

Hard balanced decoding. Every training session contains exactly 4 trials per class, so we assume the test set is balanced too. We solve $\max_\pi \sum_i \log p_{\text{ens}}(\pi(i) \mid x_i)$ subject to $|\{i : \pi(i) = c\}| = 4$ via the Hungarian algorithm. This is optimal under the balanced-class prior. A soft variant (Sinkhorn-style rescaling toward uniform) is reported in the ablation for comparison.

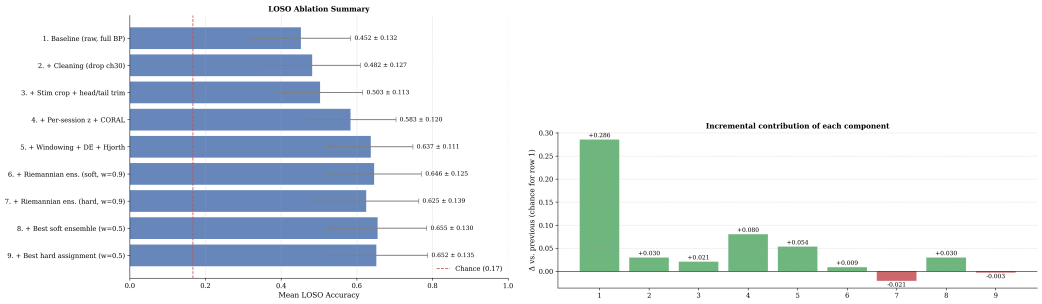
4 Results

All numbers are 14-fold LOSO accuracy (mean $\pm$ std). Chance is $1/6 = 0.167$.

Findings. *(1) CORAL is the largest single contribution* ($+0.080$, Row 3$\rightarrow$4), more than the next two components combined. *(2) The Riemannian head adds little alone* ($+0.009$ at w=0.9) but reshapes the ensemble: the optimal w=0.5 shows the two heads carry roughly equal information once calibration is accounted for. *(3) Cross-session variance dominates.* Per-row std stays at 0.11–0.14 throughout. Session 9 is consistently the worst fold (0.30–0.42) despite passing the quality

#	Pipeline	Mean	Std
1	Baseline (raw signal, full-trial band power)	0.452	0.132
2	+ Cleaning (drop ch30)	0.482	0.127
3	+ Stim crop, head/tail trim	0.503	0.113
4	+ Per-session z + CORAL	0.583	0.110
5	+ Windowed features, DE, Hjorth	0.637	0.111
6	+ Riemannian ensemble, soft, w=0.9	0.646	0.125
7	+ Riemannian ensemble, hard, w=0.9	0.639	0.139
8	Final (soft, best w=0.5)	0.655	0.130
9	**Final (hard, best w=0.5) – submitted**	**0.652**	0.135

Table 1: LOSO ablation. Row 9 (hard balanced assignment) is the delivered submission.



(a) Mean ± std per row.

(b) Incremental Δ per row (chance for Row 1).

Figure 3: LOSO ablation summary. CORAL (Row 3→4) is the single largest jump.

audit; further gains likely require addressing session-specific shift rather than refining the average pipeline.

Submission. We deliver `submission.csv` (Row 9, hard balanced assignment): exactly 4 trials per class via Hungarian assignment, optimal under the balanced-class prior.

5 Conclusion and future work

Under a strict cross-session protocol, our spectral + Riemannian ensemble with CORAL alignment reaches 0.652 ± 0.135 mean LOSO accuracy, nearly $4\times$ chance. CORAL is the single largest gain ($+0.080$), but the residual cross-session variance (± 0.13) is now the bottleneck – four times the gap between adjacent ablation rows.

Four directions would unlock further headroom: (1) *Channel montage.* Spatial priors (DASM/RASM, frontal-alpha asymmetry) and per-band Riemannian classifiers cannot be applied without electrode positions. (2) *More sessions per subject.* Reduces LOSO variance and enables session-invariant representations directly (e.g. domain-adversarial training). (3) *Tune window size and features.* Window length, hop, band edges, and the DE estimator are all heuristic; a clean grid search with proper LOSO would help. (4) *Validate emotion-onset latency.* The 3 s front-trim is a guess. Neuroscience and psychology work, or a small behavioral pilot, could pin down when each emotion actually engages.

References

[1] Ruo-Nan Duan, Jia-Yi Zhu, and Bao-Liang Lu. Differential entropy feature for eeg-based emotion classification. In *6th International IEEE/EMBS Conference on Neural Engineering (NER)*, pages 81–84, 2013.

[2] Baochuan Sun, Jiashi Feng, and Kate Saenko. Return of frustratingly easy domain adaptation. In *AAAI*, 2016.

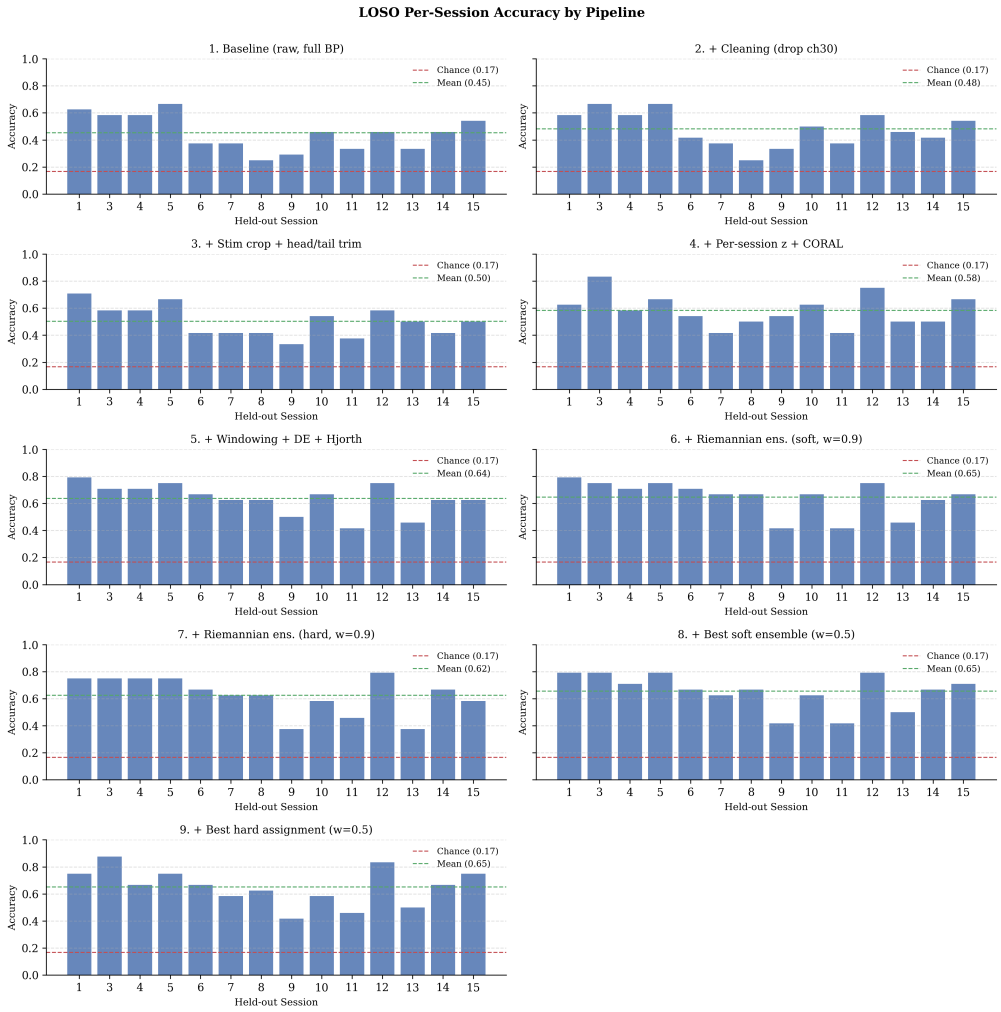


Figure 4: Per-session LOSO accuracy. Session 9 underperforms across all pipelines, suggesting a structural shift not captured by amplitude-based diagnostics.

[3] Yilun Chen, Ami Wiesel, Yonina C Eldar, and Alfred O Hero. Shrinkage algorithms for MMSE covariance estimation. *IEEE Transactions on Signal Processing*, 58(10):5016–5029, 2010.